A Tumor-to-Brain Fractal Ratio for Interpretable Brain Tumor Classification in MRI: Analysis, Validation, and Limits

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***Abstract*—Deep networks classify brain tumor type from MRI with high reported accuracy, but the features they use cannot be named, and the accuracies most often quoted are not measured at the level at which a clinical decision is made. We take the opposite position and ask how far a small set of named, measurable quantities can go. The central quantity is a ratio, $C = F_t/F_b$, between the box-counting fractal dimension of the tumor region and that of the same patient's brain with the tumor removed. Because the two regions come from one acquisition, the ratio cancels the scanner and sequence factors that make a raw fractal dimension incomparable between patients, and it expresses how irregular a tumor is relative to the brain that hosts it. Combined with fifteen further interpretable descriptors of texture, entropy, lacunarity, shape and edge sharpness, and classified with a random forest under patient-grouped cross-validation, the set reaches a patient-level balanced accuracy of 0.92 on the Figshare brain tumor dataset, against 0.913 for the benchmark published with that dataset. We also report what the descriptor does not do. Taken alone it separates pituitary adenomas well (AUC 0.907) and meningiomas barely (AUC 0.613), and 45% of its class separation survives the removal of tumor size, so it is neither a pure morphology measure nor a proxy for size. Two published methods obtain higher accuracy on the same data using learned or higher-dimensional representations. The contribution is therefore a characterisation: what a single interpretable ratio measures, how much of the task it can carry, and where it fails.**

***Index Terms*—Brain tumor classification, fractal dimension, box counting, interpretable features, magnetic resonance imaging, patient-level evaluation.**

I. Introduction

THE three most common primary intracranial tumors seen on contrast-enhanced T1-weighted MRI, meningioma, glioma and pituitary adenoma, differ in ways a radiologist describes verbally: one is well circumscribed and attached to the dura, another infiltrates surrounding parenchyma with an irregular margin, the third occupies a characteristic anatomical location. These descriptions are qualitative, and the interest in automating them is long standing [1]–[3].

Most recent progress on this task has come from transferring convolutional networks trained on natural images [3], [4] or from architectures designed to preserve spatial relationships [5]. Reported accuracies on the standard benchmark now exceed 95%. Two properties of that literature motivate the present work.

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The first is that the features are not nameable. A network that reaches high accuracy provides a decision but not a quantity a clinician can inspect, compare against a measurement made by hand, or reason about when it is wrong. Saliency maps indicate where a network attended, which is not the same as stating what it measured. In settings where a prediction must be justified rather than merely supplied, this matters.

The second is that reported accuracies are frequently not comparable to each other. Each patient in this dataset contributes many adjacent slices with almost identical appearance. A random split over slices therefore places near-duplicates in both training and test sets. Yagis *et al.* measured the size of this effect on brain MRI and found that slice-level splitting inflated accuracy by 29 to 55 points relative to subject-level splitting, and that scans with randomly assigned labels reached approximately 96% accuracy under a slice-level split against 50% under a subject-level split [7]. Those measurements were made on Alzheimer's and Parkinson's cohorts rather than on tumor data, but the mechanism is a property of how the images are organised and not of the pathology: wherever one subject contributes many adjacent slices, the same inflation is available. A number produced under the two protocols is not the same number.

This paper asks what a small set of quantities that can each be named and measured achieves under an evaluation protocol that does not inflate itself. The descriptor at the centre of it is a ratio between two fractal dimensions computed on the same slice.

Fractal analysis of brain tissue is not new. Box-counting dimensions were applied to tumor detection in MRI by Iftekharuddin *et al.* [8]. Dimensions computed over both tumorous and non-tumorous tissue have been used for glioma grading [9], and fractal measures combined with a boundary regularity ratio have been used to separate tumor types [10]. What we have not found in that literature is the use of the single ratio F_t/F_b, tumor dimension over tumor-free brain dimension, as a descriptor for three-class tumor typing, and the analysis of what such a ratio does and does not capture.

The contributions are as follows.

- We define $C = F_t/F_b$ and give the reason it is comparable across patients where a raw fractal dimension is not: both regions come from one acquisition, so the multiplicative factors introduced by scanner, sequence and noise divide out.
- We measure what the descriptor captures. Its one-vs-rest AUC is 0.907 for pituitary adenoma, 0.735 for glioma

and 0.613 for meningioma, so its strength is strongly class dependent, and we explain why the morphological heterogeneity of meningioma is the likely reason.
- We test the descriptor against the obvious alternative explanation. After removing the component predictable from tumor area, 45% of its class separation remains, so it is partly but not wholly a proxy for size.
- We report classification with sixteen interpretable features under patient-grouped cross-validation, reaching 0.92 balanced accuracy at the patient level, and we position that figure honestly against published results, two of which are higher.

II. Related Work

A. Fractal Analysis of Brain Tissue

The fractal dimension quantifies how the detail in a structure changes with the scale at which it is measured, and tissue that is structurally disorganised yields a higher value than tissue that is regular. Iftekharuddin *et al.* introduced box-counting variants for tumor detection and localisation in brain MRI [8]. More recent work computes several fractal measures over tumorous and non-tumorous regions separately for glioma grading, reporting that measures taken over apparently normal tissue also carry grading information [9]. Sánchez and Martín-Landrove combine the dimension of the tumor interface with a regularity ratio to separate gliomas from meningiomas and schwannomas [10], and treat lacunarity as a complementary measure of how gappy a region is at a given scale.

These works establish that fractal measures carry information about brain tumors. They compute the measures as separate features. The ratio used here is a normalisation, and its purpose is comparability across patients rather than the addition of a new measurement.

B. Classification on the Figshare Benchmark

The dataset published by Cheng *et al.* [1] is the standard reference for three-class tumor typing, and the accompanying method, which augments the tumor region and partitions it into rings before a bag-of-words representation, reports 91.28%.

Later work follows two directions. Engineered representations include the discrete wavelet and Gabor statistics of Ismael and Abdel-Qader [6], at 91.9%, and the hybrid PCA-normalised GIST features with a regularised extreme learning machine of Gumaei *et al.* [2], at 94.23%. Learned representations include the GoogLeNet features with a support vector machine of Deepak and Ameer [3], at 98% under patient-level five-fold cross-validation, the block-wise fine-tuned VGG19 of Swati *et al.* [4] at 94.82%, and capsule networks operating on segmented tumors and coarse boundaries [5] at 90.98%.

We take these numbers at face value where the evaluation protocol is stated, and we note where it is not, since the difference between protocols is larger than the differences between many of the methods.

C. Interpretability as a Design Constraint

The argument for named features in clinical imaging is not that they are more accurate. It is that a quantity with a name can be audited. If a model assigns a lesion to a class, a clinician can ask which measurement drove the assignment, recompute that measurement from the image, and disagree with it on grounds that can be stated. A saliency map indicates the region a network attended to but does not supply a quantity, so the same audit is not available.

This distinction matters more when the training set is small. With 233 patients a learned representation has many more free parameters than the data constrain, and the features it settles on need not correspond to anything stable. A fixed set of measurements has no such freedom: the value of lacunarity for a given mask is the same regardless of what the training set contained.

The counter-argument is equally clear and is not disputed here. Engineered features encode a prior about what matters, and where that prior is wrong the representation cannot recover. The results below show one instance of exactly that, in the class for which the descriptor is weakest.

D. Evaluation Protocol

The distinction between slice-level and patient-level evaluation is treated here as a methodological requirement rather than a detail. The quantitative basis is the leakage study of Yagis *et al.* [7] described in the introduction. We group by patient identifier throughout and report patient-level figures, and we compare only against studies that state a patient-respecting protocol.

III. Methods

A. Region Separation

Three regions are derived from each slice, and the descriptor is defined over two of them.

The *brain region* is obtained by thresholding the slice at $\max(0.25\bar{I}, 10)$ where $\bar{I}$ is the mean intensity, filling holes, applying a binary opening with two iterations, and retaining the largest connected component. This step exists to exclude the black background, which would otherwise contribute empty boxes to every box-counting estimate and bias the dimension downward by an amount that depends on how much of the field of view the head occupies.

The *tumor region* is the expert mask supplied with the dataset, intersected with the brain region so that no mask pixel lying outside the estimated brain contributes.

The *tumor-free brain* is the brain region with the tumor region removed.

We use the expert masks rather than an automatic segmentation because the descriptor is the object under study. Computing it on an automatically produced mask would confound the properties of the descriptor with the errors of the segmenter, and a change in either would be indistinguishable in the result. The consequence, which we state rather than set aside, is that the method as evaluated requires a mask at inference time; Section VI returns to this.

A slice is excluded if the tumor region contains fewer than 16 pixels or the tumor-free region fewer than 64, since a box-counting estimate over a region that small is not meaningful. On the Figshare dataset this leaves 2740 of 3064 slices, from 218 of 233 patients.

Algorithm 1 Computation of the tumor-to-brain fractal ratio

Input: slice I, expert tumor mask M
$B \leftarrow$ threshold I at $\max(0.25\bar{I}, 10)$, fill holes, open, keep largest component
$T \leftarrow M \cap B$ {tumor region}
$B_0 \leftarrow B \setminus T$ {tumor-free brain}
if $|T| < 16$ **or** $|B_0| < 64$ **then**
  reject slice
end if
for $R \in \{T, B_0\}$ **do**
  rescale intensities within R to $[0, 255]$
  for $s \in \{2, 4, \ldots, 128\}$ **do**
    $N_R(s) \leftarrow \sum_{\text{boxes } b \cap R \neq \emptyset} (\lfloor \max_b / s \rfloor - \lfloor \min_b / s \rfloor + 1)$
  end for
  $F_R \leftarrow$ slope of $\log N_R(s)$ against $\log(1/s)$
end for
return $C = F_T / F_{B_0}$

B. The Fractal Descriptor

For a region R within a slice, intensities inside R are rescaled to $[0, 255]$ using the minimum and maximum within R, so that the estimate depends on the structure of R rather than on the global intensity range of the slice. The plane is covered with boxes of side s, and for each box intersecting R the number of intensity levels spanned within that box is counted, following differential box counting. Writing $N(s)$ for the total count at scale s, the dimension is the slope of the least-squares fit

$$\log N(s) \;=\; F \cdot \log\left(\tfrac{1}{s}\right) + b, \qquad s \in \{2, 4, \ldots, 128\}. \quad (1)$$

The descriptor is the ratio of the dimension of the tumor region to that of the tumor-free brain in the same slice,

$$C \;=\; \frac{F_t}{F_b}. \quad (2)$$

The reason for the ratio is comparability. A raw fractal dimension depends on acquisition parameters, field strength, contrast administration and noise, all of which vary between patients and institutions, so F_t measured on one patient is not directly comparable to F_t measured on another. The tumor-free brain in the same slice was acquired under exactly the same conditions. If those conditions act as a common multiplicative factor on both regions, the factor divides out and C expresses how irregular the tumor is *relative to the brain that hosts it*. On the Figshare dataset C takes values in $[0.428, 1.076]$.

Algorithm 1 states the computation. Two details affect the value and are fixed for all reported results. The box sizes are powers of two up to the image side, so the fit in (1) is over eight points evenly spaced in $\log s$; using fewer points makes the slope estimate unstable on small regions, and using non-dyadic sizes introduces partial boxes at the region boundary whose treatment would need a further convention. The intensity rescaling is performed within each region separately rather than over the slice, so that F_t describes the structure of the tumor rather than its intensity relative to the rest of the image; without this, a tumor that is uniformly bright would receive a different dimension from an identical tumor in a darker slice.

TABLE I
THE SIXTEEN FEATURES. EACH IS A QUANTITY THAT CAN BE COMPUTED BY HAND FROM THE SLICE AND THE MASK.

Group	Features	Measures
Fractal	C, F_b	structural irregularity
Texture	GLCM_t, GLCM_b, ρ_t	intensity co-occurrence
Entropy	H_t	intensity heterogeneity
Lacunarity	Λ_t, Λ_b	gappiness at scale
Shape	solidity, eccentricity, extent, circularity	outline regularity
Edge	density, mean $\|\nabla\|$, border $\|\nabla\|$	boundary sharpness
Size	tumor area	extent

C. The Remaining Descriptors

The fifteen features accompanying C are computed as follows, and each is reported here in enough detail to be recomputed.

Texture. A grey-level co-occurrence matrix [11] is formed at unit offset and zero angle on intensities quantised to eight levels, computed separately over the tumor and the tumor-free brain. Contrast is taken from both regions and correlation from the tumor. Eight levels rather than the conventional sixteen or thirty-two are used because the tumor regions are small, and a finer quantisation leaves most matrix entries empty.

Entropy. The Shannon entropy of tumor intensities over a 32-bin histogram, which measures heterogeneity of intensity independently of its spatial arrangement, and so complements the co-occurrence features.

Lacunarity. Following the gliding-box definition [12], the mask is covered with boxes of side eight, the mask pixel count in each box is recorded, and lacunarity is $1+\sigma^2/\mu^2$ over those counts. A region can share a fractal dimension with another while differing in lacunarity, because the first describes how detail scales and the second how the region is distributed at one scale.

Shape. Solidity, eccentricity, extent and circularity $4\pi A/P^2$ are computed on the largest connected component of the tumor mask. These describe the outline rather than the interior and are the features expected to separate a well-circumscribed lesion from an infiltrating one.

Edge. The intensity gradient magnitude is computed by finite differences. Edge density is the fraction of tumor pixels whose gradient exceeds the slice mean; mean gradient is taken over the tumor; border gradient is taken over the one-pixel dilation minus erosion of the mask, and is intended to capture how abruptly the lesion terminates.

Size. Tumor pixel count, included explicitly so that its contribution is visible and can be controlled for, as Section IV-B does.

D. The Feature Set

C alone is insufficient, for reasons the results make explicit. It is combined with fifteen further descriptors, listed in Table I, each chosen because it measures something the others do not.

The groups are complementary in a specific sense. A fractal dimension and a lacunarity can coincide for regions that differ visibly, because the first measures how detail scales and the

second how the region is distributed at a fixed scale. Shape features describe the outline rather than the interior, which is what distinguishes a well-circumscribed meningioma from an infiltrating glioma. Edge features describe the transition at the boundary, which reflects infiltration into surrounding tissue. Tumor area is included explicitly rather than left implicit, so that its contribution is visible to the analyst and can be controlled for, as Section IV-B does.

E. Classification and Evaluation

A random forest with 300 trees is used, with maximum depth and minimum leaf size selected by grid search. The evaluation protocol has three properties that together prevent the inflation described in the introduction.

Folds are formed with `GroupKFold` over five splits, with the patient identifier as the grouping variable, so that every slice from a patient falls on the same side of every split. The inner grid search likewise uses `GroupKFold` over the training patients, so hyperparameter selection cannot leak across patients either. Slice predictions are aggregated to one prediction per patient by majority vote, and the reported figure is balanced accuracy computed over patients.

Balanced accuracy, the unweighted mean of per-class recall, is used rather than plain accuracy because the classes are uneven: 708 meningioma, 1426 glioma and 930 pituitary slices. A classifier that predicts glioma for every input attains 46.5% plain accuracy while carrying no diagnostic information; the same classifier attains 0.333 balanced accuracy.

F. Implementation and Cost

Feature extraction requires one pass over each slice and is dominated by the two box-counting estimates, which are $O(HW)$ per scale over eight scales. On the 2740 usable slices the full extraction takes approximately six minutes on a single GPU-equipped Kaggle instance, with no GPU operations involved; the cost is in the mask operations and the box counting, both of which run on the CPU. Model fitting with the nested grid search adds approximately ten minutes.

The complete pipeline, the extracted features and the per-fold results are available in the repository accompanying this paper, together with the seed used for every split, so that each reported figure can be recomputed rather than taken on trust.

IV. Results

A. What the Descriptor Separates

Fig. 1(a) reports the one-vs-rest AUC of C used alone as a single-feature classifier.

The values are 0.907 for pituitary adenoma, 0.735 for glioma and 0.613 for meningioma. A single ratio therefore carries most of the information needed to identify one class and little of what is needed to identify another.

The pattern is consistent with what the ratio measures. Pituitary adenomas are compact and comparatively homogeneous, so the dimension of the tumor region sits well below that of the surrounding brain and C is low. Gliomas are irregular and heterogeneous, raising F_t towards or above F_b. Meningiomas

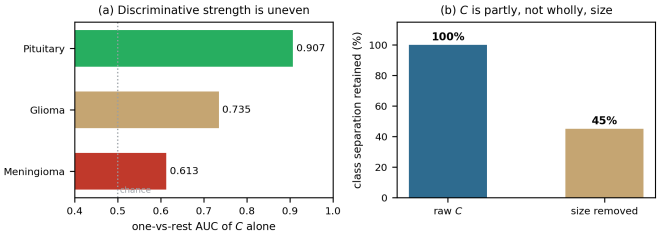


Fig. 1. (a) One-vs-rest AUC of C alone. The descriptor separates pituitary adenomas well and meningiomas barely. (b) Class separation retained after removing the component of C predictable from tumor area.

TABLE II
PUBLISHED RESULTS ON THE FIGSHARE DATASET. THE PROTOCOL COLUMN RECORDS WHETHER THE EVALUATION GROUPS BY PATIENT, SINCE FIGURES OBTAINED UNDER DIFFERENT PROTOCOLS ARE NOT DIRECTLY COMPARABLE.

Method	Representation	Protocol	Acc.
Cheng 2015 [1]	BoW, ring partition	patient	0.913
Ismael 2018 [6]	DWT + Gabor	split	0.919
This work	**16 named features**	**patient**	**0.920**
Gumaei 2019 [2]	PCA-NGIST + RELM	5-fold	0.942
Swati 2019 [4]	VGG19 fine-tuned	patient	0.948
Deepak 2019 [3]	GoogLeNet + SVM	patient	0.980

fall between the two and, more importantly, vary widely among themselves: some are smooth and well circumscribed while others are lobulated with irregular margins. A single scalar summarising irregularity cannot separate a class whose defining property is not a consistent degree of irregularity.

We report this rather than only the aggregate model performance because it delimits the claim. C is a useful descriptor, not a sufficient one, and the class for which it is weakest is identified.

B. Is the Descriptor Measuring Size?

Fractal estimates are sensitive to the area of the region measured, and the three tumor types differ in typical size, so a descriptor that appears to capture morphology could be capturing size. The alternative explanation has to be tested rather than dismissed.

We removed from C the component predictable from tumor area and recomputed the class separation. Fig. 1(b) shows that 45% of the separation remains. The descriptor is therefore partly confounded with size and partly independent of it. Both halves of that statement are part of the result: a claim that C measures morphology alone would be wrong, and a claim that it is merely size in another form would also be wrong.

C. Classification

Under the protocol of Section III-E, the sixteen-feature model reaches a patient-level balanced accuracy of 0.92. Table II places this against published results on the same dataset.

The result exceeds the benchmark published with the dataset by less than one point, which we do not treat as a meaningful margin. Two later methods are higher: Gumaei *et al.* by 2.2

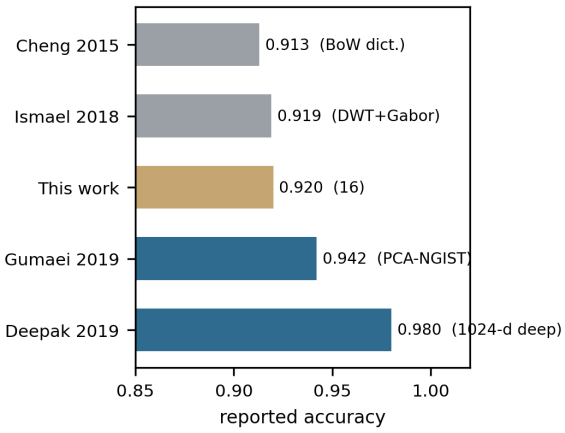


Fig. 2. Reported accuracy against the size and nature of the representation. The present result is competitive with the benchmark published alongside the dataset and below two later methods that use higher-dimensional or learned features.

points using a hybrid feature representation, and Deepak and Ameer by 6 points using transferred deep features under a patient-level protocol.

We state this plainly because the alternative invites a reader to make the comparison for us. The claim of this paper is not that sixteen interpretable features outperform a transferred convolutional network. It is that they come within six points of one while remaining individually nameable, measurable and open to inspection, which is a different property and useful for a different reason.

V. Discussion

Three observations follow from the results.

The descriptor behaves as its construction predicts. C was defined to measure irregularity relative to the host brain, and it separates most strongly the class that is most consistently regular relative to its surroundings. It separates weakly the class whose members vary most among themselves. A descriptor that performed uniformly across all three would in fact be harder to interpret, since its behaviour would not follow from what it measures.

The size analysis marks the boundary of the morphological claim. Retaining 45% of separation after removing size means the descriptor carries information that tumor area does not, and also that a substantial part of its apparent discriminative power is area in another form. For a descriptor intended to be interpretable this is worth knowing precisely, because an interpretation that attributes the whole effect to structural irregularity would be wrong.

The comparison with the literature is a trade rather than a defeat. The gap to the strongest reported method is six points of accuracy. What is obtained in exchange is a representation in which every dimension has a name and a computation: a clinician can ask why a slice was assigned to a class and receive an answer in terms of measured lacunarity, boundary gradient and the fractal ratio, each of which can be checked against the image. Whether that exchange is worth making depends on the setting, and we do not claim it is always worth making.

Finally, the setting in which this trade is most defensible is worth stating explicitly, since a general claim would be too strong. Three conditions make a named-feature representation preferable: the training cohort is small enough that a learned representation is weakly constrained; a mask or region of interest is available for other reasons, so computing region-based descriptors adds little cost; and the prediction must be explained to someone who will act on it. All three hold in the setting studied here. Where a large cohort is available and the prediction is used for triage rather than justification, the argument runs the other way, and the accuracy difference in Table II is the relevant quantity.

VI. Limitations

A mask is required at inference. The descriptor is a ratio between two regions, so those regions must be delineated before it can be computed. We used expert masks in order to isolate the descriptor from segmentation error. A deployed system would need an automatic region extraction step, and the accuracy reported here would not transfer unchanged to that setting; the sensitivity of C to mask error is not characterised in this work and would need to be measured directly.

The descriptor is weak for meningioma. An AUC of 0.613 is close enough to chance that C should not be relied on for that class. The full feature set compensates, but a reader interested in the descriptor itself should take this as its principal limitation.

Partial size confounding. 45% of separation survives the removal of size, which means 55% does not.

Single dataset, single modality. All results are on contrast-enhanced T1-weighted slices from one collection. Fractal estimates depend on resolution and reconstruction, so transfer to other acquisition protocols is not established here.

Two published methods are more accurate. Stated in Table II and repeated here so that it is not overlooked.

The novelty claim is bounded. Fractal analysis of brain tumors, and the computation of dimensions over both tumorous and non-tumorous tissue, are established. What we have not found in the literature is this particular ratio used as a descriptor for three-class typing. A literature search establishes absence of evidence rather than evidence of absence, and the claim is phrased accordingly.

VII. Conclusion

A single ratio between the fractal dimension of a tumor and that of the brain tissue surrounding it is comparable across patients, because both quantities come from one acquisition and the factors that differ between acquisitions divide out. Used alone it identifies pituitary adenomas well and meningiomas poorly, and roughly half of its discriminative power is attributable to tumor size. Combined with fifteen further named descriptors it classifies tumor type at 0.92 balanced accuracy at the patient level, close to the benchmark published with the dataset and below the strongest transferred deep representation. The value of the approach lies in what can

be said about a prediction rather than in the accuracy of the prediction itself, and the analysis above is intended to make clear both what the descriptor measures and where it stops being useful.

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